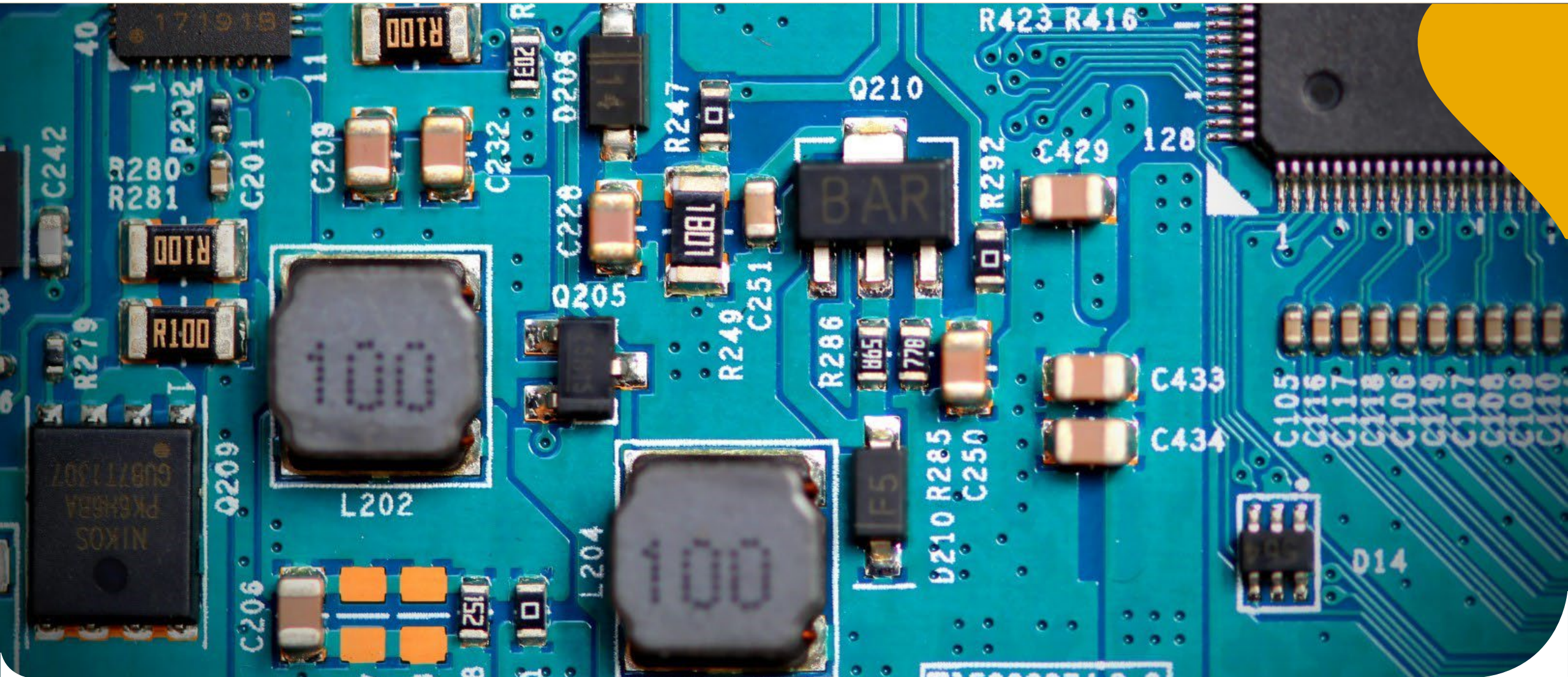


Design(E)-314

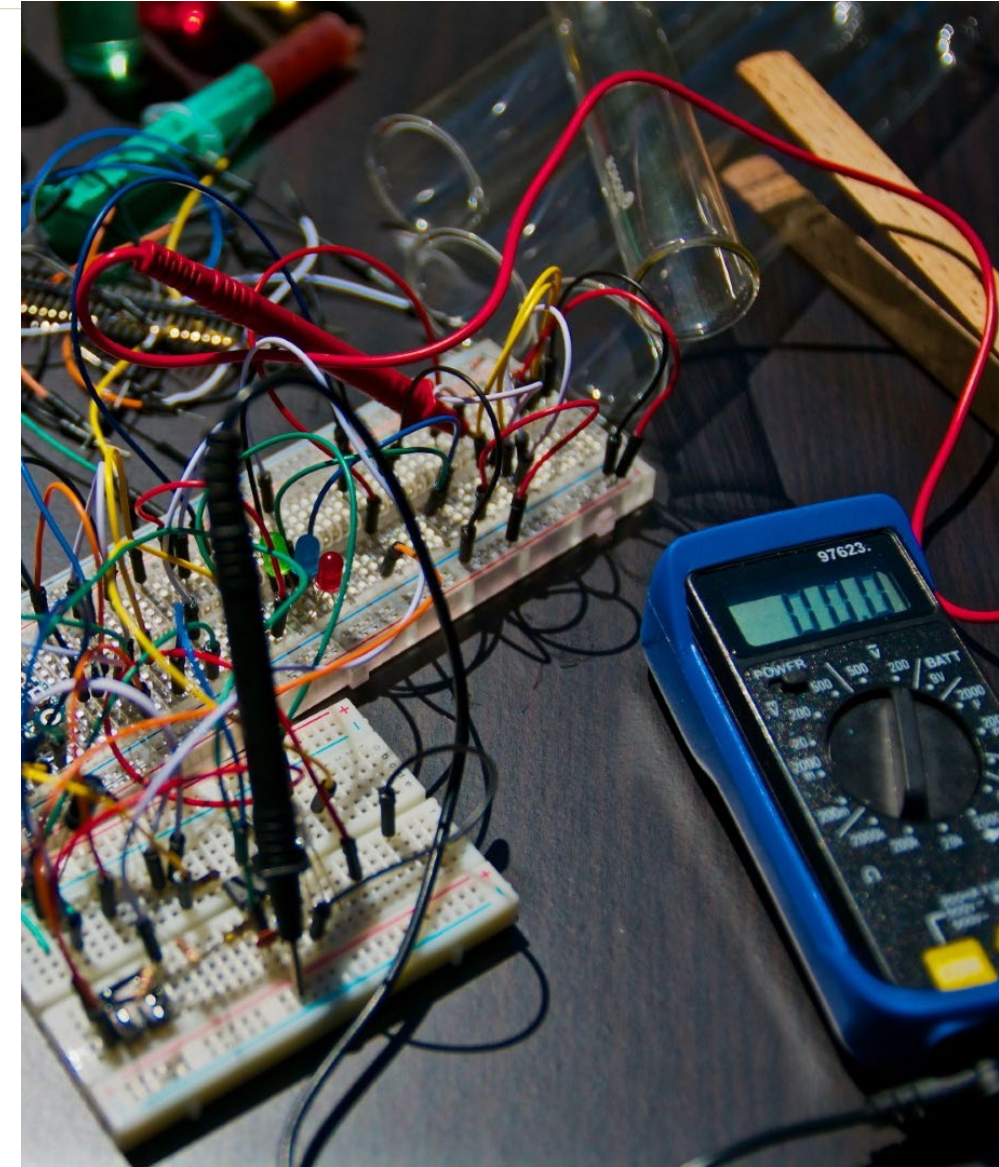
Lecture 5



Overview

We are in the Demo 3 cycle

- Test 2 – Review
- Demo 3 cycle
 - Light sensor (L4)
 - Op-Amp (L4)
 - ADC (L4)
 - Keypad interface
 - LCD/OLED
 - UART command interface
 - State machine handling
- Prelim Report



PDD - Updates

- Version 0.7 is the latest version

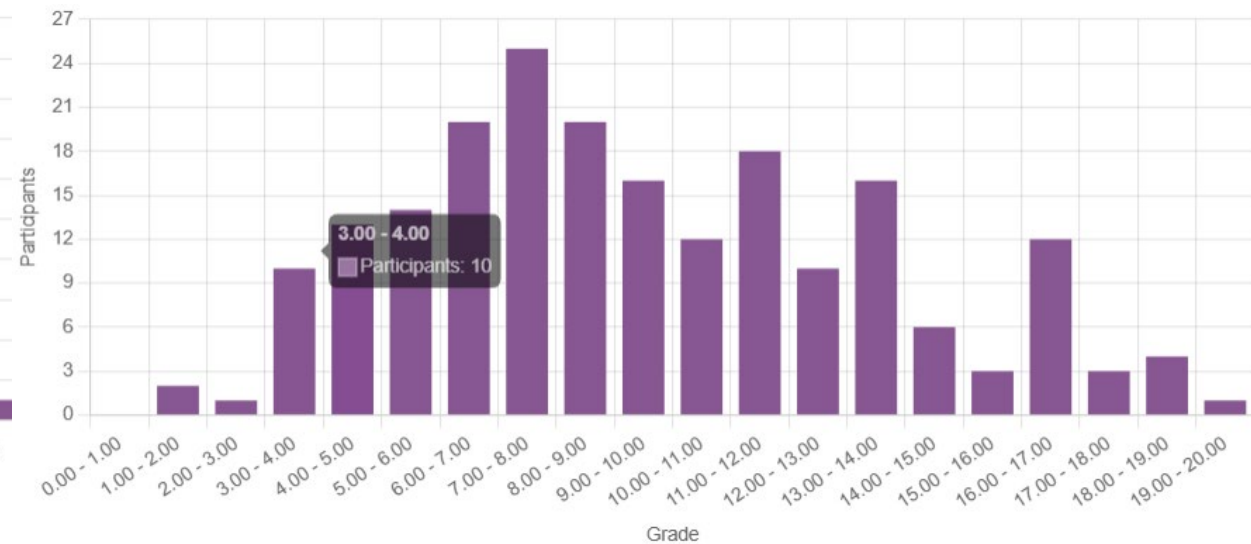
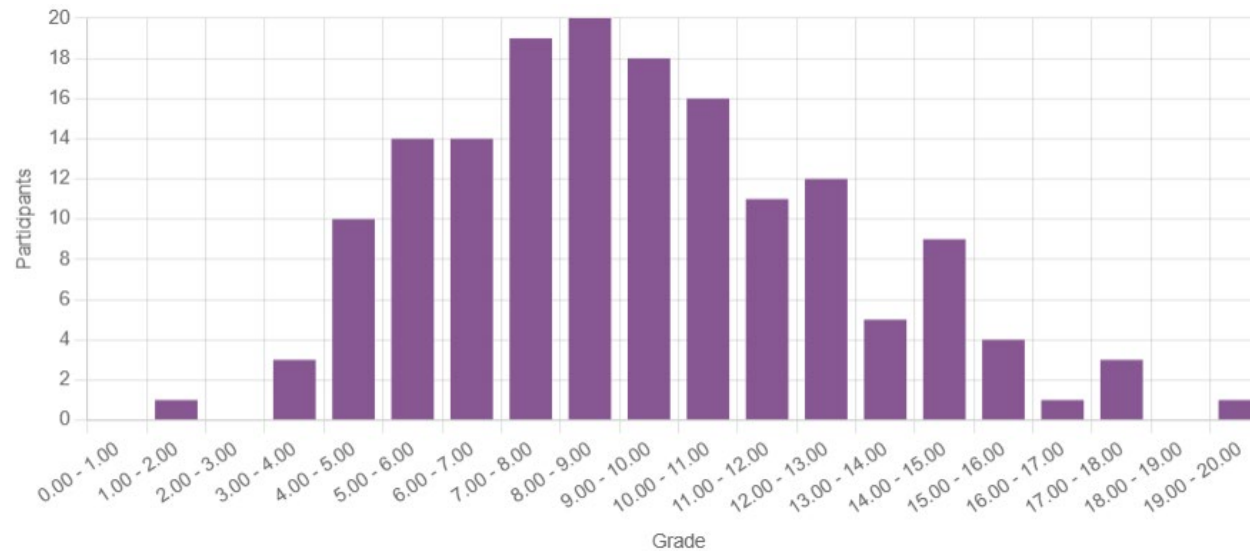
Version	Date	Changes
0.1	9 Feb 2026	- Initial project specification and guide
0.2	11 Feb 2026	- Added Demo 1 information in Tables 10 and 11
0.3	19 Feb 2026	- Clearly specified the Demonstration 1 UART message format for the temperature sensor in Tables 10.
0.4	3 Mar 2026	- Added Demo 2 information in Tables 10 and 11 - Added information related to preliminary report
0.5	5 Mar 2026	- Added clarification to checking of alarm conditions, and SB response to Stat command.
0.6	27 Mar 2026	- Added information pertinent to Demo 3.
0.7	2 Apr 2026	- Added information regarding Light sensor calibration in Section 3.8.

- Major change in v0.7: Specification for Demo 2 and preliminary report

Test 2

Review and Feedback

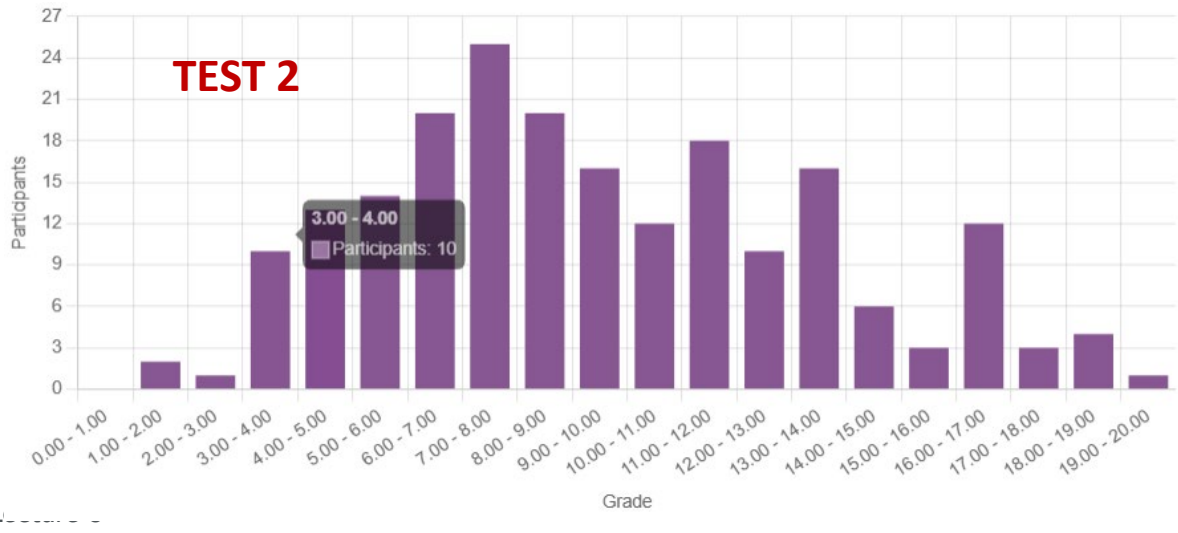
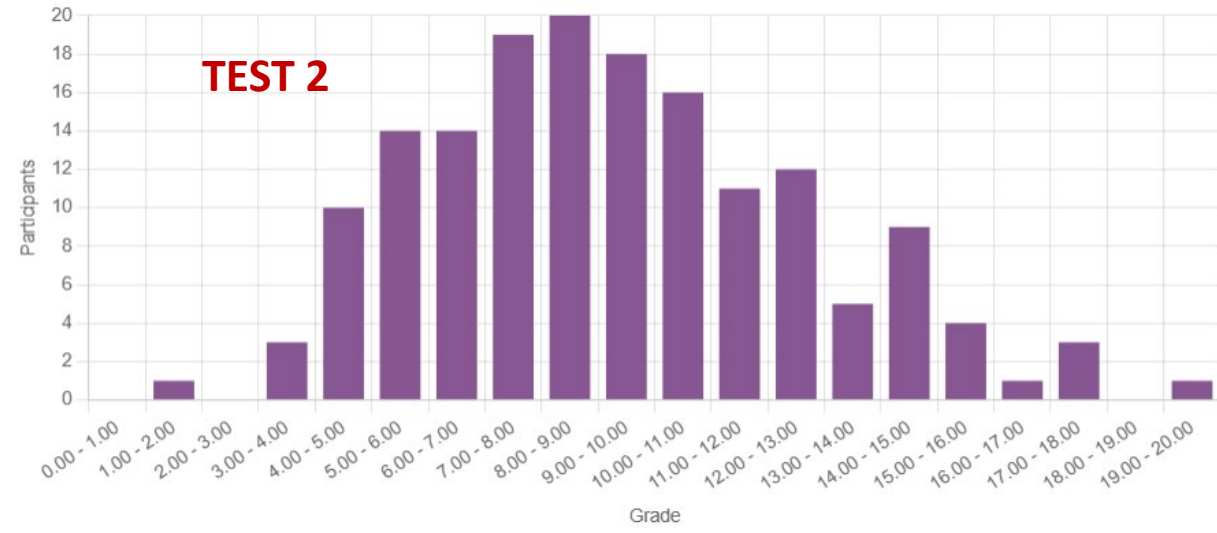
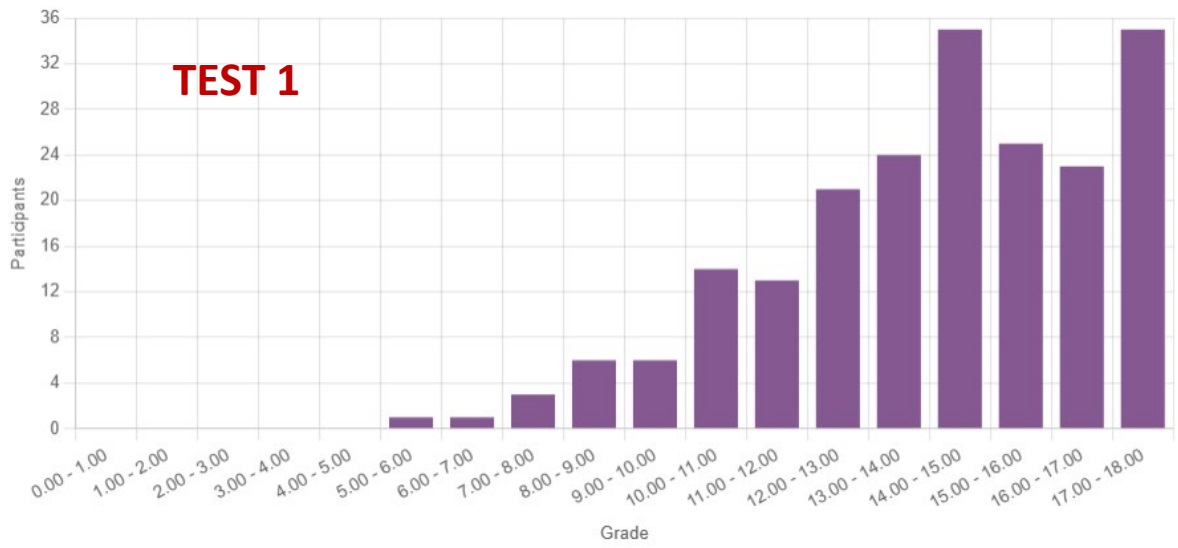
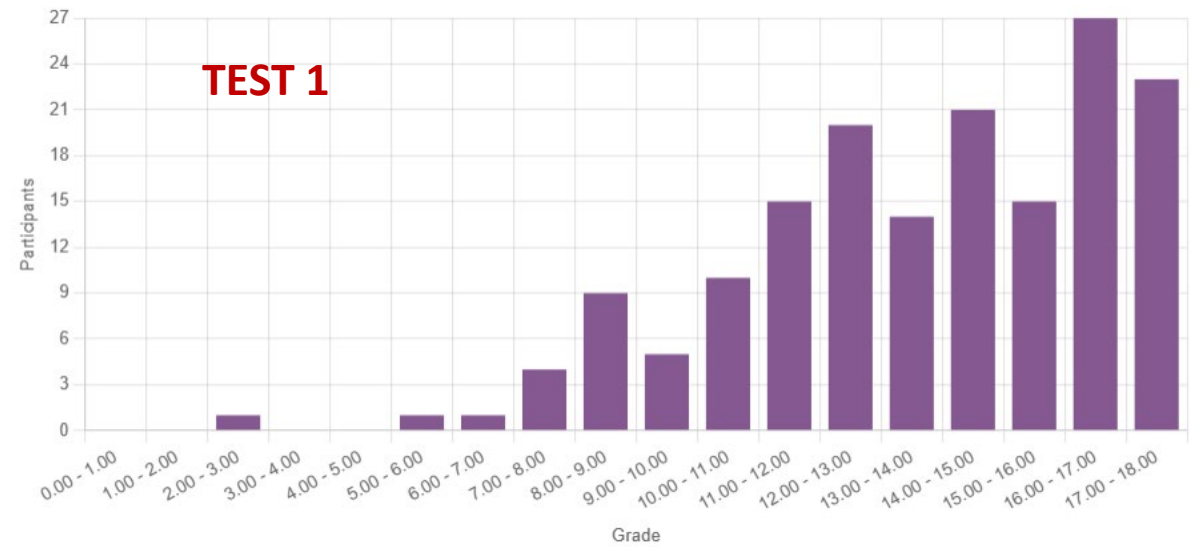
Results



Avg 45% - 47%

Test 2

Review and Feedback



Test 2

Review and Feedback



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Based on the HC-SR204 timing diagram, which indicates minimum and maximum ECHO pulse widths, what is the minimum distance that the device can report back on the ECHO pin? // Gebaseer op die HC-SR204 tydsdiagram, wat minimum en maksimum ECHO-pulswydtes aandui, wat is die minimum afstand wat die toestel op die ECHO-pen kan terugrapporteer?

Min Pulse Width = cm

Max Pulse Width = cm

Answer:

- Datasheet = 150us minimum pulse width.
- $150\text{us} \times 0.034\text{cm/us} = 5.1\text{cm}$ but need to divide by 2
- Answer = $5.1/2 = 2.55\text{cm}$

Answer:

- Datasheet = 25ms max pulse width.
- $25\text{ms} \times 34\text{cm/ms}$ divide by 2
- Answer = 425cm

Test 2

Review and Feedback



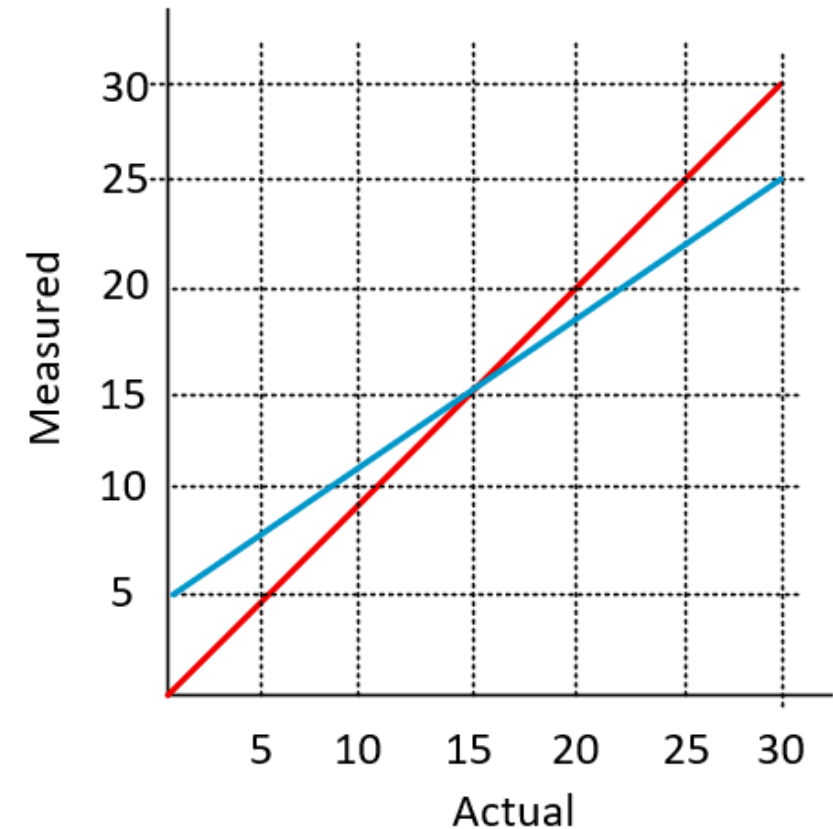
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Ultrasonic Calibration (Q3: class avg. 34%)

You have measured your Ultra sonic sensor's output values. And the results are plotted below (blue line). The true values are plotted in red as a perfect linear equation. You decide to calibrate the raw measurements (in blue). What should your calibration coefficients be if you apply a linear regression straight line formula of $y = \alpha x + \beta$. Please provide the values for alpha α and beta β .

Answer: alpha = $\frac{3}{2}$ and $\beta = -5$



Test 2

Review and Feedback



Ultrasonic Resolution (class avg. 20%)

a) Your friend has decided to implement the HC-SR04 ultrasonic sensor. On the ECHO pin, you receive a pulse. You make use of an External Interrupt to read the rising and falling interrupt pulses from the ultrasonic sensor, which allow you to determine the length of the Echo pulse in terms of time. You have decided to make use of the HAL_GetTick() function to get the time stamp of the start and stop of the Echo pin's pulse. What is the maximum measurement resolution you will have in terms of distance (cm). Assume the speed of sound is 340m/s

** resolution is dependent on sampling time of HAL_GetTick*
of samples = $\frac{\text{speed}}{\text{sample } \Delta t}$
sample time = 1ms

Answer

- Speed of sound = 340m/s = 34000cm/s x 1s/1000ms = 34cm/ms
- Thus, answer is 17 cm *→ The signal travels there and back*

Test 2

Review and Feedback



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Ultrasonic Resolution (class avg. 20%)

b) Should you use TIM2 as external Timer of the STM32 to capture the pulse width of the ECHO pin, what is the maximum resolution you can measure in terms of distance? Assume the speed of sound is 340m/s. You have set up the external timer at a clock speed of 1 MHz.

Answer

- Speed of sound = 340m/s = 34000cm/s $\times 1\text{s}/10000000\text{us} = 0.034\text{cm}$
- Thus, answer is 0.017 cm

Test 2

Review and Feedback

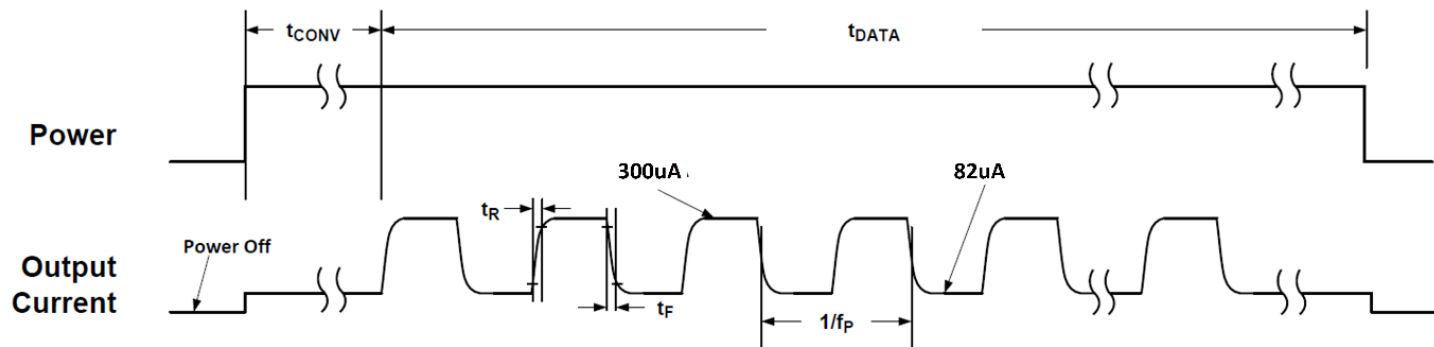


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LMT01 Resistors (class avg. 44%)

You have been provided with an LMT01 temp sensor. For your EDesign project your GPIO pins have a minimum V_{IH} of $V_{IH} = 0.7V_{DD}$ and a maximum V_{IL} of $0.3V_{DD}$. Assume V_{DD} is 3.3V. You have received an LMT01 sensor, which is exactly the same as the one used for your EDesign project. However, the only difference is that the output current of the LMT01 datasheet is as in the image below. Select **all the resistor values** that you could use for this LMT01 sensor so that you satisfy the V_{IH} and V_{IL} requirements of the STM32's GPIO pins.



Answer:

- $82\mu A * R_x \leq 0.99$
- $300\mu A * R_x \geq 2.31$
- **Result:** $7700 \leq R_x \leq 12073.2$

Light Sensor Operational Amplifier

Op-Amp Design

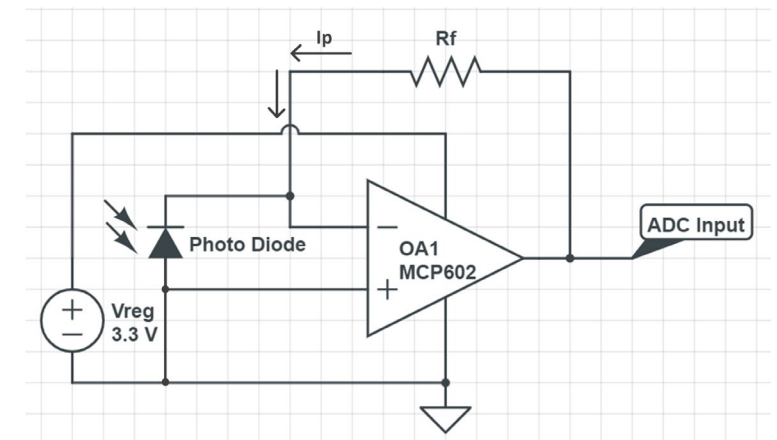
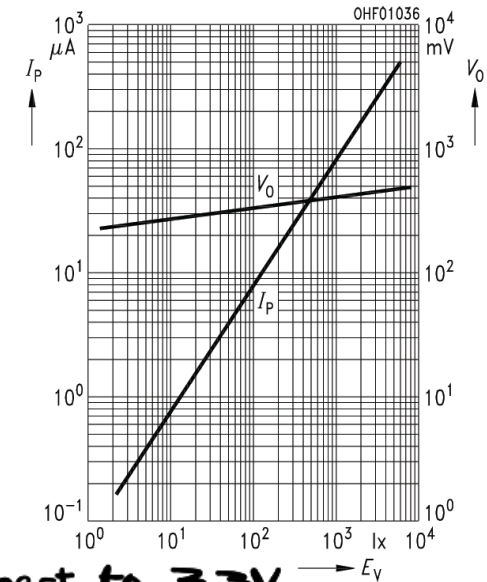
- Use MCP602 Op-Amp for your circuit design
- Op-Amp Design provided

Calibration

- You must calibrate your photo-diode using the White LED that was provided to you.
- Ensure a high level of accuracy 0 – 1000 Lux
- Design the board to provide 3.3V at a maximum of 1000 LUX
- Allows you to take advantage of the full ADC resolution (0 – 3.3V)

Design for saturation

connect to 3.3V



Light Sensor Operational Amplifier

Calibration

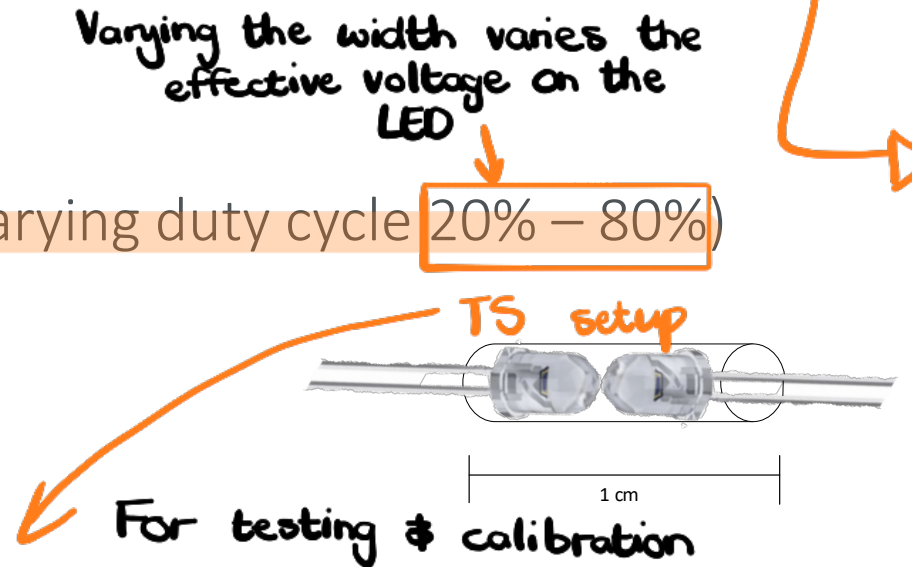
- Use the calibration table to ensure that your own readings are accurate

Calibration constants for the photodiode LUX circuit.

Setup

- signal generator: 2.5Vpp
- Square wave PWM signal (varying duty cycle 20% – 80%)
- 100 Hz
- DC offset: 1.3V

Setup	
Vpp	2.5Vpp
DC Offset	1.3Vdc
Hz	100 Hz
Calibration	
Duty Cycle [%]	Lux
20	260
30	390
40	520
50	660
60	780
70	900
80	1060



Physical setup

- LED and light-sensor placed on top of each other
- Place within cylindrically 3x folded paper 1cm length
- Dark/Covered surrounding

Keypad

Function

- The keypad serves as a tool to enter information
- The keypad also serves as a tool to navigate menu states/options



Keypad



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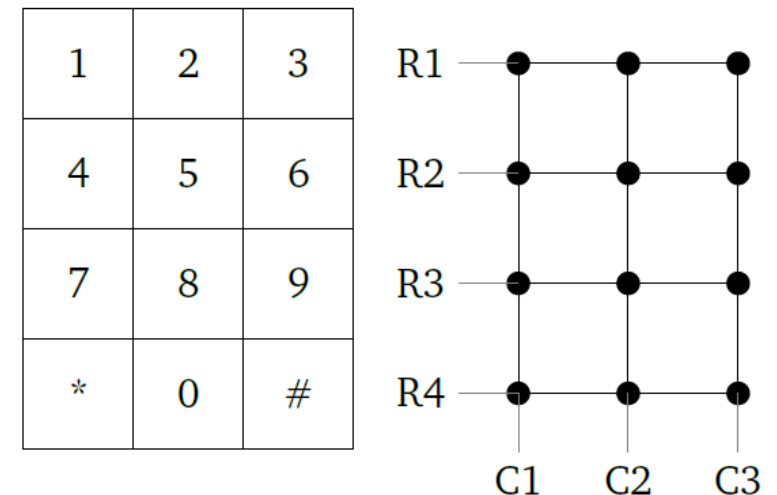
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Hardware interfacing

- The keypad has 4x rows and 3x columns
- 12 keys; 7x GPIO pin connections required to STM32
- Treat each key as a button interface
- You can use an active high/low config (use high)
- When no key is pressed, the input should read low.

- **Note:** The rows and columns are intermixed
- The PCB pin order from left to right is:

- COL2, ROW1, COL1, ROW4, COL3, ROW2, ROW1



Demo 3 Cycle

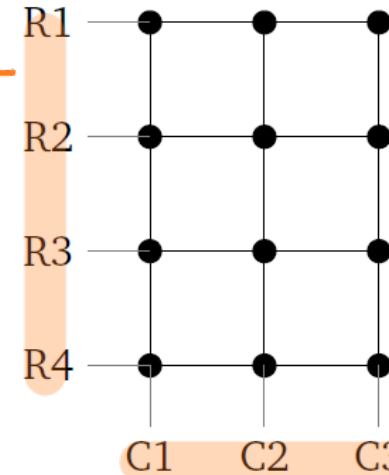
Recommended configuration

- Rows = outputs
- Columns = interrupt driven inputs (use internal or external pull-down)

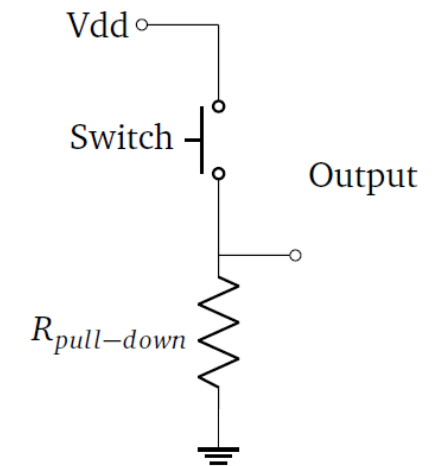
Keypad setup: Active high keypad configuration

- Must connect the rows and columns in 'active high' configuration
- When a key is pressed, a high input is read.

connect the rows to
GPIO-Output
(Acts as the V_{DD})



Connect columns to
GPIO_EXT



Pro-Tip:

use the internally available pull-up/pulldown resistors to simplify circuit setup



Keypad



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Software interfacing

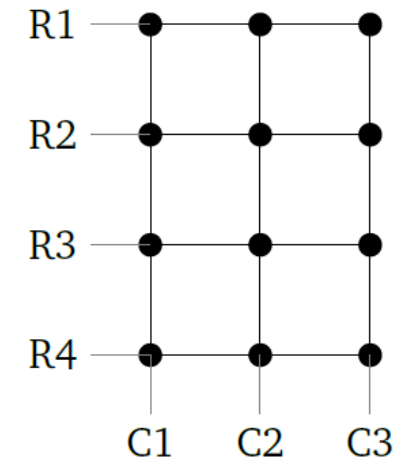
- Debouncing is required
- Assign the rows or columns (not both) as external interrupts

Class brain-storm

- How will you ensure that you can capture the moment when a key is pressed and know exactly which key has been pressed?

↳ When the column flag goes high & low 

1	2	3
4	5	6
7	8	9
*	0	#



Pro-Tip:

- Wait for release (to avoid repeats)
- 1 press = 1x press and 1x release



OLED



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- Objective: to present useful information to the user.
- I²C interfacing (use only one of the STM32 I2C channels)
- Resolution: 128 x 32 individual blue OLED pixels
- Input voltage: 2.5V-5.5V
- Graphic Driver IC is SSD1306
 - Therefore, to design your software driver, you should refer to the SSD1306 datasheet.
 - The datasheet contains important information regarding the interface and command options available.



*** NB**



Pro-Tip:

- drivers available online (<https://github.com/afiskon/stm32-ssd1306>)
- recommended that you use the 7x10 pixel font

OLED

- The OLED display must support at least **18x3 characters** : 3 rows; 18 characters per row
- Your driver-level software should also manage the timing control to update the display information to the OLED/LCD.
- This functionality is usually not explicitly included in online driver libraries.
- Menu interface to be discussed in Lecture 6

Table 5: OLED/LCD display table for the user interface.

Menu Level	Description	Row	Display OLED
1	display page default page	1	=YYYY/MM/DD HH:MM=
		2	== Measurements ==
		3	Press -> display
2	data entry page	1	=YYYY/MM/DD HH:MM=
		2	=== Data Entry ===
		3	Press -> display
3	diagnostics page	1	=YYYY/MM/DD HH:MM=
		2	== Diagnostics ==
		3	Press -> display



OLED State Machine



- As you enter into different menus, some of the features on your board will have dual functionality
- For example, S3 button can either set/reset warning conditions, or it can be used to confirm data entry
- Its functionality depends on its state

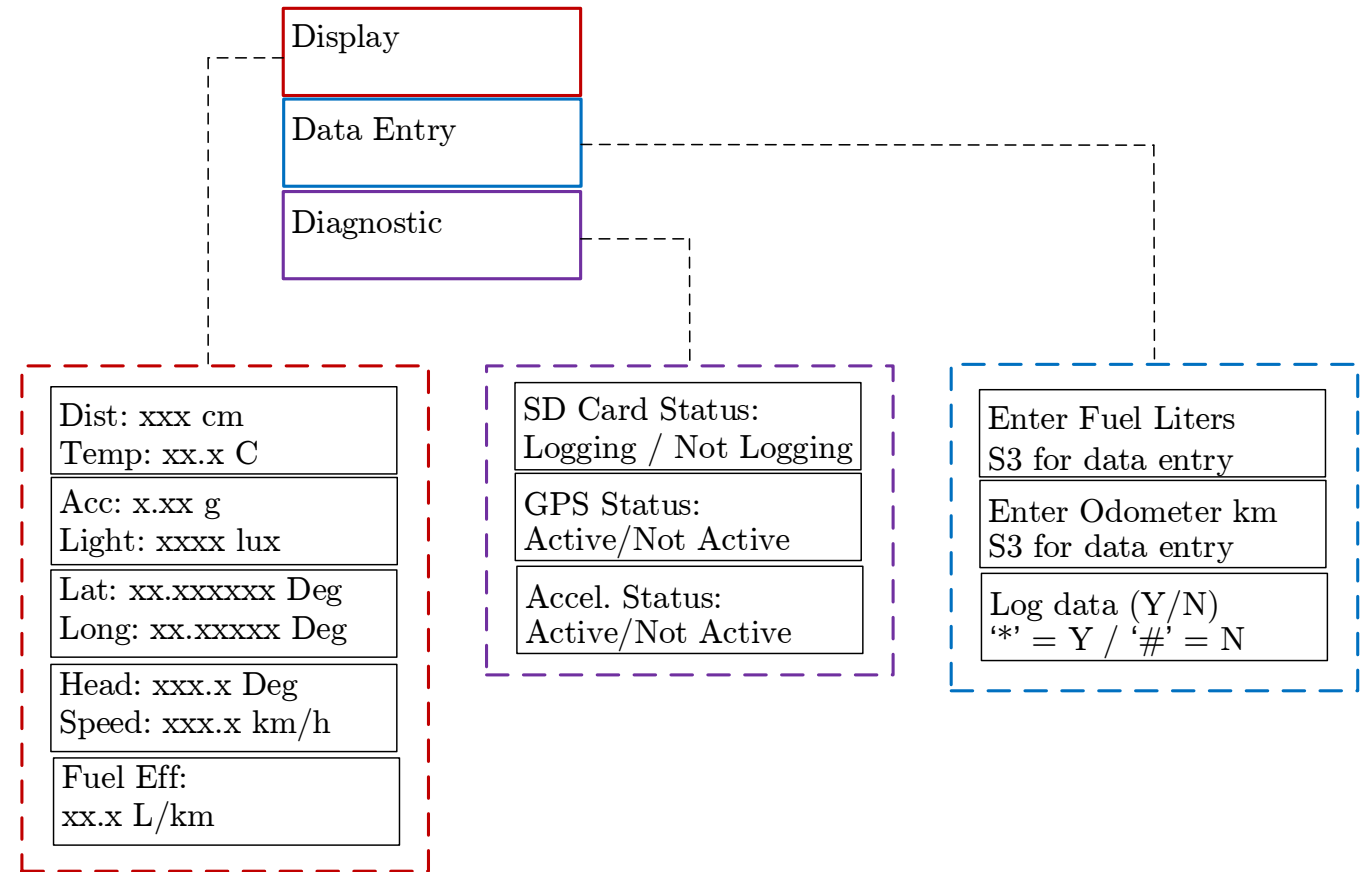
Table 5: OLED/LCD display table for the user interface.

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		2	=== Data Entry ===
		3	Press -> display
3	diagnostics page	1	=YYYY/MM/DD HH:MM=
		2	== Diagnostics ==
		3	Press -> display



OLED State Machine

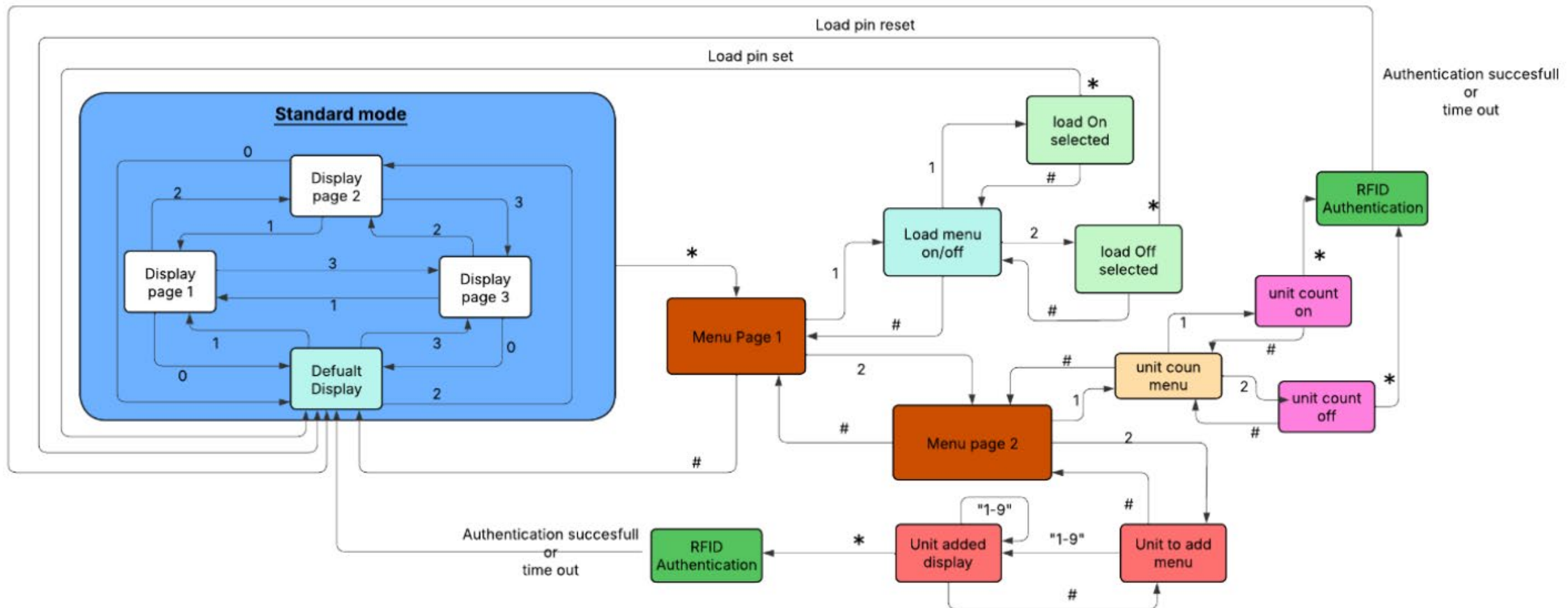
- It is important to draw your own menu-interfacing diagram and state machine to keep track of where you are.



Note: This is a fictional/example OLED display state machine

OLED State Machine

- Good example of a menu interfacing and OLED display State Machine



UART Command Interface



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Command format	Description
@Stat&\n	Request SB status information
@Log&\n	Toggle logging to SD Card ON/OFF
@Dump&\n	Dump SD card file contents over UART to PC/TS
@SetWarn C=Y&\n	Set Warning Condition 'C' to condition 'Y' (Y=1 to set; Y=0 to reset)
@SFD xxx.x;zzz.z&\n	Sending Fuel Data (Fuel fill-up of xxx.x liter, zzz.z Vehicle Kilometers)
@RFE&\n	Request Fuel Efficiency
@CLF&\n	Clear File to erase SD card data
@SetRTC YYYY/MM/DD;HH:MM:SS&\n	Set the RTC to the specified date 'YYYY/MM/DD' and time 'HH:MM:SS'

- The UART Command Responses/Actions are listed in the PDD
- Commands are from the PC (or TS) to your SB.

UART Command Interface



- The UART Command Responses/Actions are listed in the PDD
- These are defined as from the SB to the PC (or TS).

Command received	Response by SB
None - at Start-up	*12345678#\n
Stat	@YYYY/MM/DD HH:MM:SS \n Distance: xx.x cm\n Temperature: ±xx.x C\n Light: xxxx lux\n X accel: ±x.xx g\n Y accel: ±x.xx g\n Z accel: ±x.xx g\n Unsafe driving: y\n Impact detected: y\n Low-Light warning: y\n Proximity warning: y\n High Temperature: y\n GPSLat: ±xx.xxxxxx\n GPSLong: ±xxx.xxxxxx\n&\n
Dump	@ followed by Line-by-line information as stored in the .csv file. Terminated by &\n
SetWarn	No UART response
SFD	No UART response
RFE	@YYYY/MM/DD HH:MM:SS\n Fuel Eff: xx.x km/L xx.x L/100km\n&\n
CLF	No UART response
SetRTC	No UART response

UART Command Interface



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SetWarn command: @SetWarn C=Y&\n

- The SetWarn-command serves to override the sensor measurements by setting or resetting warning conditions. An integer value of 'Y=1' is used for setting the condition and an integer value of 'Y=0' for resetting the condition. 'C' is a integer value from 1 to 5 which specifies the warning condition.

- C=1 'Unsafe driving' warning condition
- C=2 'Impact detected' warning condition
- C=3 'Low-light' warning condition
- C=4 'Proximity' warning condition
- C=5 'Uncomfortable temperature' warning condition



Demo 3:

TS will Set Impact Detected Warning Condition with 'SetWarn conditions' and verify that status-indicating **LED** is flashing and that **OLED** displays warning



S3: disables warning condition

UART Command Interface

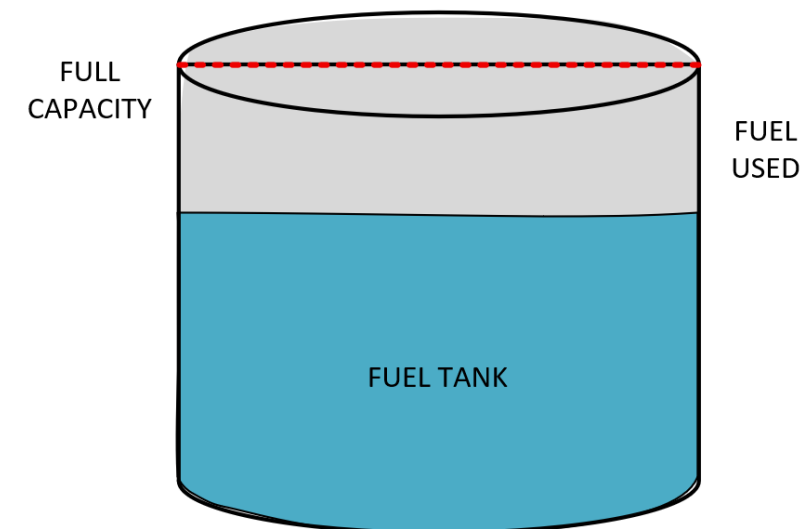


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SFD command: @SFD xxx.x;zzz.z&\n

- The SFD-command serves to set the amount litres refuelled (when refuelling at a service station) as well as the distance travelled (in kilometers) since the previous refuelling.
- The command format of SFD xxx.x;zzz.z indicates that xxx.x litre of fuel has been refuelled while a distance of zzz.z km has been travelled since the previous refuelling.
- It is assumed that the vehicle fuel tank is refuelled to full capacity each time, so that we have a consistent reference point.



UART Command Interface

RFE command: @RFE&\n

- The RFE-command is used to calculate and return the fuel efficiency.
- It is assumed that the SFD-command has been previously received and executed.
- The system must return the fuel efficiency in both kilometre per litre (km/L) and litre per 100km (L/100km).

Demo 2 Cycle Preliminary Report



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Preliminary report

- Submission closes: **7 April 2026 @ 12:00** (mid-day)

Peer-review process

- Each student will peer-review 3x other reports
- Use the rubric provided to assess the designs.
- Provide sensible feedback to your peers regarding their report / designs and what they can do to improve their report / designs in future.
- Peer-review due date: **16 April 2026 @ 12:00** (mid-day)

Pro-Tip:

- Follow the rubric to the best of your ability
- Your ability to review a design weighs equal to your ability to communicate your design

Marking/Weights

- 50%*obtained mark + 50%*peer-review-scoring-ability

Questions?

